1. **Dynamic Programming** (25%)

A Markov Decision Process (MDP) with three states {A, B, C} is presented in Figure 1. The actions, state transitions and rewards are also shown in the figure. For example, when action i is taken in state A, the reward is +6 and the system state transits to C or back to A with probability ¾ and ¼, respectively. If action j is taken while in state A, the reward is -4 and the state is changed to B with probability 1. The discount factor is γ. The return for any given state s is the total of discounted rewards over time starting from state s by following an action policy π.

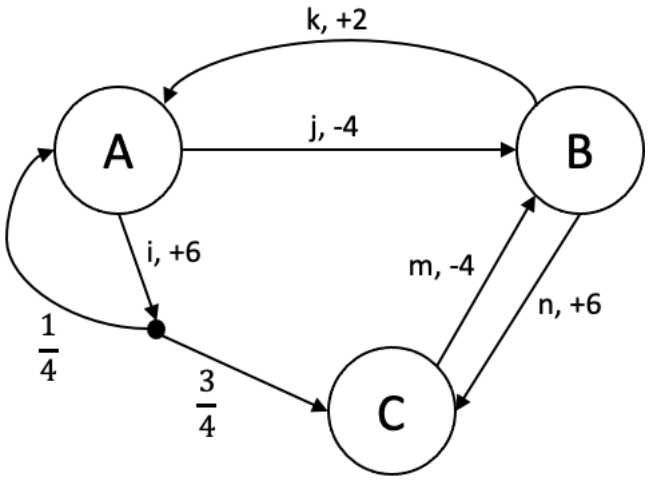


Figure 1. A simple Markov Decision Process (MDP)

(a) Construct a table to represent the above MDP. Let s, a, s', r and $p(s'|s,a)$ denote the current state, action, next state, reward and transition probability from s to s' given current state s and action a taken. The table should have columns for s, a, s', r and $p(s'|s,a)$, and one row for every 4-tupe of (s,a,s',r) for which $p(s'|s,a) > 0$. [5]

Answer:

s	a	s'	r	$p(s'\|s,a)$
A	i	C	+6	3/4
A	i	A	+6	1/4
A	j	B	-4	1
B	n	C	+6	1
B	k	A	+2	1
C	m	B	-4	1

(b) Give the Bellman optimality equation for $q_*(s,a)$ when state $s = A$ and action $a = i$. To expand one time step is sufficient. [4]

Answer:

$$q_*(A,i) = \frac{1}{4} \times \left[6 + \gamma \max_{i,j} \left(q_*(A,i),\ q_*(A,j)\right)\right] + \frac{3}{4}[6 + \gamma q_*(C,m)]$$

(c) Consider the uniform random policy $\pi(s,a)$ that takes all possible actions from the current state with equal probability. For example, $\pi(A,i) = \pi(A,j) = 0.5$. Initialize the state value to be 0 for all states. Perform one synchronous round of policy evaluation iteration. [5]

Answer:

$$v(\text{A}) = 0.5 \times 6 + 0.5 \times -4 = 1,$$

$$v(\text{B}) = 0.5 \times 2 + 0.5 \times 6 = 4,$$

$$v(\text{C}) = -4$$

(d) Using the updated state-value function from part (c), apply one iteration of greedy policy improvement to obtain an "optimal" deterministic policy $\pi_g(s)$ for each state $s = A, B$ or C. Use $\gamma = 0.9$. [5]

Answer: Since

$$q(A,i) = 6 + 0.9 \times \left(\frac{1}{4} \times v(A) + \frac{3}{4} \times v(C)\right) = 3.525,$$

$$q(A,j) = -4 + 0.9 \times v(B) = -0.4,$$

$$q(B,n) = 6 + 0.9 \times v(C) = 2.4 \text{ and}$$

$$q(B,k) = 2 + 0.9 \times v(A) = 2.9,$$

$$\pi_g(A) = i,\ \pi_g(B) = k,\ \pi_g(C) = m.$$

(e) Initialize the state value to be 0 for all states. Perform two synchronous rounds of value iteration and fill in the following table with the appropriate values. Use $\gamma = 0.9$. [6]

State	$q(s,a)$					Best action	$v(s) = \max_a q(s,a)$
	i	j	k	m	n		
A							
B							
C							
A							
B							
C							

Answer:

State	$q(s,a)$					Best action	$v(s) = \max_a q(s,a)$
	i	j	k	m	n		
A	6	-4	-	-	-	i	6
B	-	-	2	-	6	n	6

C	-	-	-	-4	-	m	-4
A	4.65	1.4	-	-	-	i	4.65
B	-	-	7.4	-	2.4	k	7.4
C	-	-	-	-4	-	m	-4

2. Batch TD(0) Learning (25%)

Consider an undiscounted ($\gamma = 1$) Markov Reward Process (MRP) with two states A and B. The transition matrix and reward function are unknown, but three sample episodes have been observed:

(i) A+4 → A+3 → B−4 → A+3 → B−3 → terminate

(ii) B−1 → A+3 → B−2 → terminate

(iii) B−2 → A+3 → A+2 → terminate

In the above episodes, state transitions and rewards are shown at each step. For example, A+3 → A in the last episode (iii) indicates a transition from state A to state A with a reward of +3.

(a) Use the first-visit Monte-Carlo (MC) method to estimate the state-value functions of V(A) and V(B). [5]

Answer:

V(A) = [(4+3-4+3-3) + (3-2) + (3+2)]/3 = 3,

V(B) = [(-4+3-3) + (-1+3-2) + (-2+3+2)]/3 = -0.33

(b) Use the every-visit MC method to estimate the state-value functions of V(A) and V(B). [5]

Answer:

V(A) = [(4+3-4+3-3) + (3-4+3-3) + (3-3) + (3-2) + (3+2) + 2]/6 = 1.67

V(B) = [(-4+3-3) + (-3) + (-1+3-2) + (-2) + (-2+3+2)]/5 = -1.2

(c) Assume that the three episodes are repeatedly used in the learning process. As a result, the episodes sufficiently capture the transition probabilities between states A, B and the terminal state. Obtain these state transition probabilities. [5]

Answer:

$p(A \rightarrow B)$ = 3/6 = 0.5,

$p(A \rightarrow A)$ = 2/6 = 0.33,

$p(A \rightarrow \text{Terminate})$ = 1/6 = 0.17,

$p(B \rightarrow A)$ = 3/5 = 0.6,

$p(B \rightarrow \text{Terminate})$ = 2/5 = 0.4

(d) What are the average rewards associated with state A and B, as reflected in the three episodes? [5]

Answer: The average reward for state A is (4+3+3+3+3+2)/6=3. Similarly, the average reward for state B is (-4-3-1-2-2)/5=-2.4

(e) Using the state transition probabilities and the average state rewards in parts (c) and (d), give the Bellman expectation equations for the MRP. Solve the values of V(A) and V(B) for the MRP. [5]

Answer: The Bellman expectation equation for state A is

$$V(A) = E(R_A) + \gamma[P(A \to A)V(A) + P(A \to B)V(B)] = 3 + 0.33\,V(A) + 0.5\,V(B).$$

Similarly, the Bellman equation for state B is

$$V(B) = E(R_B) + \gamma[P(B \to A)V(A) + P(B \to B)V(B)] = -2.4 + 0.6\,V(A).$$

Solving these equations yields $V(A) = 4.86$ and $V(B) = 0.52$.

3. Q-Learning and Sarsa (20%)

Consider a system with four states {A, B, C, D}, two possible actions {i, j} and rewards as Figure 2 shows. For example, when action i is taken in state A, the reward is -4, and the state changes from A to B with probability 1.

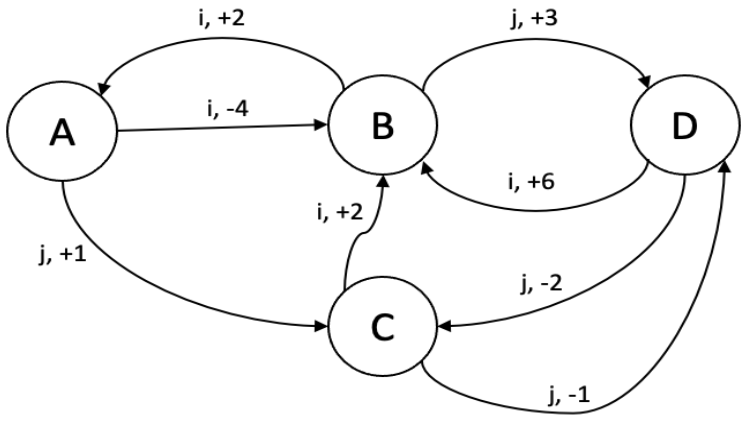


Figure 2: A four-state system.

(a) Explain the main difference between Q-learning algorithm and Sarsa algorithm. [4]

Answer: The main difference is how the target value is calculated when they update Q values. In Sarsa, the action used in the target value is selected by following the current policy. On the other hand, the action that can achieve the highest Q value will be selected to calculate the target value in Q-learning. Alternatively, it is also acceptable to say that Sarsa updates Q values using the Bellman expectation equation, while Q-learning uses Bellman optimality equation.

(b) Use the Q-learning algorithm to update Q values with the discount factor of $\gamma = 0.9$ and the learning rate of $\alpha = 0.5$. The Q values for all states and actions are initialized to 0. Consider that the system begins in state A and takes a sequence of four actions {i, j, j, i}, and the associated action-value functions are updated according to the Q-learning algorithm. Fill the following table of Q values after these updates. [8]

Q(s,a)	A	B	C	D
i				
j				

Answer: 2 marks for each correct answer in the table.

Q(s,a)	A	B	C	D
i	-2	0	1.68	0
j	0	1.5	0	-1

(c) Assume that Sarsa algorithm is used to update Q values for the same setting and the system also takes the sequence of actions {i, j, j, i} from state A, as in part (b). Fill the following table of Q values after these updates. Assume that action i is taken after the sequence of 4 actions and this final transition is not used to update Q values. [8]

Q(s,a)	A	B	C	D
i				
j				

Answer: 2 marks for each correct answer in the table.

Q(s,a)	A	B	C	D
i	-2	0	1	0
j	0	1.5	0	-1

4. **RL Controller for Base-Station Transmissions** (30%)

Consider two neighboring cells, *A* and *B*, with a base-station (BS) transmitter in each cell in Figure 3. By using frequency bandwidth of W Hz, the two BS transmitters simultaneously transmit data at the same power level P to two users, one associated with each of the cells. Let G_A and G_B be the large set of users associated with the respective cells.

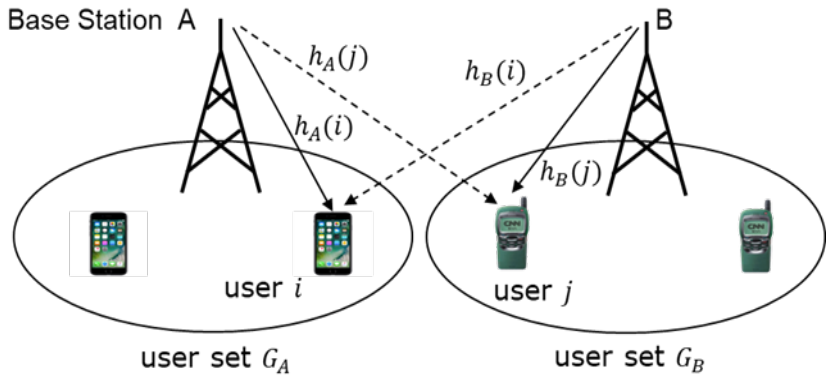


Figure 3. Downlink transmissions in two neighboring cells A and B.

Let system time be divided into *time slots* and 8 consecutive time slots form a *time frame* in Figure 4. Both BS transmitters simultaneously send data in each time slot to two users indexed i and j where $i \in G_A$ and $j \in G_B$.

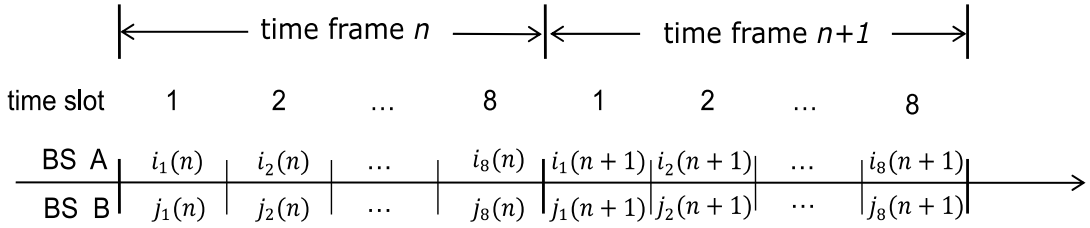


Figure 4. Transmission schedule in the time slots and time frame.

Due to radio interference (with noise neglected), the signal-to-interference ratio (SIR) α_i at user $i \in G_A$ or G_B is given by

$$\alpha_i = \frac{h_A(i)\,P}{h_B(i)\,P} \quad \text{or} \quad \alpha_i = \frac{h_B(i)\,P}{h_A(i)P}\ ,$$

respectively, where P is the transmission power, and $h_A(i)$ and $h_B(i)$, which are greater than 0, denote the corresponding channel gain from BS transmitter A and B to user i. Assume that $h_A(i)$ and $h_B(i)$ for all user i can be accurately estimated in advance and do not change when the users are receiving data from the BS's over a number of time frames. By the Shannon capacity formula, data rate R_i for user $i \in G_A$ or G_B is

$$R_i = \mathrm{W}\log_2\alpha_i = \mathrm{W}\log_2\left(\frac{h_A(i)}{h_B(i)}\right) \quad \text{or} \quad R_i = W\log_2\alpha_i = \mathrm{W}\log_2\left(\frac{h_B(i)}{h_A(i)}\right),$$

respectively. With probability σ, the concurrent transmissions from both BS transmitters in each time slot will continue in the same time slot in the next time frame. With probability $1-\sigma$, the transmissions in a time slot will complete and the same time slot in the next time frame onward will become available for transmission to two new users, one from G_A and another from G_B.

Consider that there are k time slots available for $k = 0, 1, \ldots, 8$ at the end of a time frame. We aim to design an RL controller (agent) to perform the pairing of k new users from G_A with another k

new users from G_B so that one user from G_A is paired with another user from G_B to receive data in each of the k newly available time slots starting in the next time frame. As the respective channel gains, $h_A(i)$ and $h_B(i)$, from the BS transmitters to each user i are known in advance to the RL controller, the objective of the user pairing is to minimize the radio interference due to simultaneous transmissions and thus maximize the total data rate of all continuing and new data transmissions in all time slots of each time frame, which is discounted over time frames with a discount factor γ.

Figure 4 shows the users in both cells scheduled to receive data transmitted from the BS transmitters in different time slots and time frames. For example, BS transmitter A and B send data to users $i_2(n) \in G_A$ and $j_2(n) \in G_B$, respectively, in time slot 2 during time frame n in the figure. If the BS transmissions in time slot m for $m = 1$ to 8 continue in time frame $n + 1$, $i_m(n + 1) = i_m(n)$ and $j_m(n + 1) = j_m(n)$. Use the notation in Figure 4 to answer the following questions.

(a) Define the system state S_n at the end of each time frame n for the RL controller. [7]

Answer: $\mathrm{S_n} = \begin{pmatrix} i_1(n) & j_1(n) \\ i_2(n) & j_2(n) \\ \vdots & \vdots \\ i_8(n) & j_8(n) \end{pmatrix}$ where $i_m(n)$ and $j_m(n)$, which are the indices of users receiving data in time slot m, are replaced by 0 (or any invalid user index) for $m = 1$ to 8 to indicate that time slot m becomes available for transmissions to new users if the original pair of transmissions in time slot m has completed during time frame n.

(b) Explain why the system with your state definition satisfies the Markovian property. [4]

Answer: (i) Transmissions of both BS in each time slot will complete with probability σ in each time frame, independent of the transmission history of the same time slot during previous time frames. (ii) Non-zero $i_m(n)$ and $j_m(n)$ in time slot $m = 1, 2, \dots, 8$ mean that both BS transmitters will continue to transmit to the same two users in the same time slot during the next time frame. (iii) Zero entries of $i_m(n)$ and $j_m(n)$ in time slot $m = 1, 2, \dots, 8$ indicate that each of the corresponding time slot now becomes available for data transmissions to one randomly chosen user from G_A and another from G_B starting from the next time frame. Such info in the state description is sufficient for predicting the evolution of the system for the next time frame.

(c) Define the actions of the RL controller at the end of each time frame. [5]

Answer: For each time slot $m = 1, 2, \dots, 8$ with non-zero $i_m(n)$ and $j_m(n)$ in S_n, the corresponding transmissions from the BS transmitters will continue. Therefore, there is no action for these time slots.

By identifying $i_m(n)$ and $j_m(n)$ in S_n being zero for time slot $m = 1, 2, \ldots, 8$, determine the number of available time slots for transmissions to new users to be k, starting in the next time frame.

Randomly select k new users from G_A and another k users from G_B to receive data in the identified k time slots. For each of the k news users from G_A, pair up with one of the k users from G_B to receive data from the BS transmitters in one of the remaining available time slots, starting from next time frame. (Precisely which available time slot for a given pair of new users does not matter and the controller can randomly select one from the remaining available time slots.)

(d) Define the reward for each time frame n, including the new transmissions in some time slots, to achieve the objective stated above. [5]

Answer: The channel gains are known to the RL controller because it knows the users who are receiving data in various time slots in both cells in each time frame. The (immediate) reward for each time frame n is the total data rate of data transmissions in all 8 time slots during the time frame, which is given by

$$R_n = W \left[\sum_{m=1}^{8} \log_2 \left(\frac{h_A(i_m(n))}{h_B(i_m(n))} \right) + \log_2 \left(\frac{h_B(j_m(n))}{h_A(j_m(n))} \right) \right]$$

(e) Suggest one RL algorithm to perform the pairing of users in real time for simultaneous reception of data from the BS transmitters in each available time slot and explain why it is appropriate. [4]

Answer: Deep Q-learning and actor-critic RL approaches are applicable to solve the formulated problem. It is so because both approaches support real-time management of the user pairing.

(e) To highlight the performance of your RL approach to pairing up users for receiving data from the BS transmitters, suggest a simple, baseline scheme for such comparison. Explain why your suggested scheme is appropriate. [5]

Answer: Random pairing of the $2k$ new users for receiving data if there is k available time slots in the next time frame. That is, each of the k users in G_A is paired up with one randomly chosen user among the k users from G_B to receive data from two BS transmitters in the same time slot. Such random pairing does not consider the path gains or data rates of the simultaneous transmissions by avoiding mutual interference in the same time slot. In contrast, the total data rate by proper user pairing will be optimized in the learning process of the RL algorithm.